

MANMOHAN SINGH

Software Engineer & CS Student

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PROFESSIONAL SUMMARY

Computer Science (AIML) student with production-level experience in *Agentic AI, Full-Stack Development, and ML Deployment*. Successfully architected and shipped *4+ end-to-end projects*, specializing in autonomous multi-agent systems.

EDUCATION

GLA University | Mathura, India

Expected June 2027

Bachelor of Technology (BTech) in Computer Science (Spec. in AIML and IoT)

PROJECTS

ARS – Autonomous Research Scientist

Stack: MERN, FastAPI, LangGraph, CrewAI, Groq API, RAG, Agentic AI, Docker, vector DB

- **Architected a multi-agent research system** utilizing **StateGraph** and **10 specialized nodes** (Search, Reasoning, Coder) to automate the end-to-end research lifecycle, effectively reducing manual effort by **70%**.
- **Engineered a triple layered validation framework** - comprising Peer Review, Code Critic, and Paper Critic loops to ensure rigorous quality control and achieve a **84%+ research validity rate**.
- **Developed an Optimized hybrid SLM (Llama-3.1-8b) and LLM (Llama-3.3-70b) pipeline** via **Ollama, MLX and Groq API**, integrating **Docker** for remote sandboxing to ensure secure, high-performance code execution.

Attendify – Smart Attendance System

Stack: React.js, Node.js, MongoDB, Google Maps API, Fast API, YOLO library, Firebase

- **Architected a fraud resistant, full-stack attendance system** utilizing **MERN & FastAPI**, ensuring secure data persistence and a responsive user interface for real-time tracking.
- **Engineered a multi-factor authentication (MFA) framework** by integrating **Geofencing (GPS)** and **Facial Recognition** technologies, significantly enhancing identity verification and location accuracy.
- **Reduced proxy attendance by 85%** by implementing **virtual boundaries** and real-time monitoring, streamlining administrative oversight and ensuring integrity in employee and student attendance records.

Restmage – AI Map Builder & Cost Estimator

Stack: React.js, Node.js, Express.js, MongoDB, Tailwind, Gemini API, RAG, AWS

- **Architected a scalable MongoDB, Express.js, React.js and Node.js platform on AWS** to automate high-fidelity architectural floorplan generation, modernizing design workflows through secure, cloud-native infrastructure.
- **Integrated Gemini API for generative architectural design** and engineered a data-driven cost estimation engine, bridging the gap between intelligent planning and real-time financial forecasting.
- **Engineered cost-forecasting ML models** achieving R^2 of 0.89 across 12 architectural parameters, significantly optimizing budget reliability and resource allocation efficiency

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL, HTML, CSS

Frameworks: React.js, Next.js, Node.js, Express.js, Tailwind CSS, Fast API

Databases: MongoDB, MySQL, Vector DB (FAISS/Pinecone)

DevOps & Cloud: Docker, Kubernetes, Git, AWS, Linux, Postman, Firebase

AI/ML: PyTorch, Scikit-learn, LLMs, RAG, Agentic AI, LangChain, LangGraph, CrewAI, Gemini/Groq API

CERTIFICATIONS & ACHIEVEMENTS

- Meta Front-End Developer Professional Certificate | Coursera (2025)
- Cisco thingQbator: Full Stack MERN Development | Cisco (2024)
- Python for Data Science & AI | Infosys Springboard (2024)
- Team Leader | Smart India Hackathon (SIH) (2023)
- Web Developer | Cyberonites Club - Built official club portal (cyberonites.com) (2025)